

ECON 0100 | Fall 2024

MiniExam E | Version 1 | Rubric

Question 1 | Profit Maximizing Quantity (15 pts)

Answer: 24 units

- Full credit (15 pts): 24 units
- Partial credit:
 - 13 pts: Shows $MR=MC$ work with calculation error
 - 5 pts: Solves for something but not $MR=MC$
- Zero: No work shown

Question 2: Pre-Entry Price (15 pts)

Answer: \$126

- Full credit (15 pts): \$126
- Partial credit:
 - 15 pts: Right method, wrong quantity from Q1
 - 13 pts: Correct demand curve use with calculation error
 - 5 pts: Uses MR curve or other mistake
- Zero: No work shown

Question 3: Pre-Entry DWL (15 pts)

Answer: $\frac{1}{2} \cdot (30 - 24) \cdot (126 - 102) = 72$ with $Q_{SE} = 30$

- Full credit (15 pts): 72
- Partial credit:
 - 13 pts: Triangle method with correct Q_{SE} but calculation error
 - 8 pts: Triangle method with incorrect Q_{SE}
 - 8 pts: Correct Q_{SE} but incorrect DWL idea
 - 3 pts: Did something but incorrect
- Zero: Blank

Question 4: Total Quantity After Entry (20 pts)

Answer: $20 + 20 = 40$ units

Note: If students answer 20 here, we'll give them 18 points. We'll check their understanding of how to use market quantity in question 5.

- Full credit (20 pts): 40 units
- Partial credit:
 - 18 pts: Solves symmetric duopoly with minor calculation error
 - 14 pts: Incorrect approach to NE but uses $MC = MR$ to find BR
 - 8 pts: Incorrect approach and doesn't use $MC = MR$
 - 3 pts: Does something but no BR
- Zero: Blank

Question 5: Post-Entry Price (20 pts)

Answer: $\$110 = 150 - 20 - 20$

- Full credit (20 pts): \$110
- Partial credit:
 - 15 pts: Uses correct demand curve with calculation error
 - 8 pts: Uses demand curve but uses $Q = 20$
 - 8 pts: Uses $Q = 40$ incorrectly
 - 4 pts: Shows some reasoning about competition's effect but incorrect
- Zero: Blank

Question 6: Long-Run Competitive DWL (15 pts)

Answer: 0

- Full credit (15 pts): Zero/No deadweight loss
- Partial credit:
 - 10 pts: Explains competition eliminates DWL, unclear reasoning
 - 5 pts: Recognizes price approaches MC but wrong conclusion
 - 3 pts: Fails to understand perfect competition
- Zero: Blank

Common Mistakes to Watch For

1. Using MR curve instead of demand curve for pricing
2. Forgetting to double quantity for duopoly total
3. Assuming price stays same after entry

4. Calculating DWL without consideration of market structure
5. Missing the symmetry in duopoly market